

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

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Introduction

Mean Squared Error (MSE) is one of the most commonly used performance metrics for regression problems. It measures the average squared difference between the actual values and the predicted values. A lower MSE indicates better prediction accuracy.

In this problem, the neural network achieves a **training MSE of 0.85**, which is very low, but the **testing MSE is 5.20**, which is much higher. This large difference indicates that the model performs very well on the training data but fails to generalize to unseen data. This situation usually points to **overfitting**, where the model memorizes the training data instead of learning general patterns.

Analysis of the Difference Between Training and Testing MSE

1. Overfitting

The most common reason for the large difference between training and testing MSE is overfitting.

- The neural network has learned the training data extremely well.
- It may have memorized specific patterns and noise present in the training dataset.
- When new testing data is provided, the model cannot accurately predict the outputs because it has not learned generalized relationships.
- A training MSE of **0.85** and testing MSE of **5.20** strongly indicate poor generalization.

Reasoning:

The model fits the training examples too closely, resulting in low training error but high testing error.

2. Model Complexity

The neural network may contain too many hidden layers or too many neurons.

- A highly complex model has a large number of trainable parameters.
- Such models can easily memorize the training dataset.
- Instead of learning meaningful relationships, they learn unnecessary details.

Reasoning:

Higher complexity increases variance, making the model sensitive to training data.

3. Insufficient Training Data

If only a limited amount of student performance data is available,

- the neural network does not observe enough examples,
- making it difficult to learn generalized patterns.

Small datasets increase the chance of overfitting because the model repeatedly sees the same examples.

Reasoning:

More training data generally improves generalization ability.

4. Presence of Noise in Training Data

Training data may include

- incorrect marks,
- missing values,
- data entry errors,
- outliers.

The neural network may learn these noisy observations instead of actual relationships.

Reasoning:

Noise increases training accuracy artificially while reducing testing performance.

5. Poor Feature Selection

Student score prediction depends on relevant features such as

- attendance,
- assignment marks,
- previous semester GPA,
- study hours,
- internal assessments.

If irrelevant or redundant features are included,

- the model becomes confused,
- prediction accuracy decreases.

Reasoning:

Poor-quality features reduce the model's ability to generalize.

6. Data Distribution Difference

The training dataset and testing dataset should represent the same population.

If

- training data contains mostly high-performing students,
- testing data contains average or low-performing students,

the model performs poorly.

Reasoning:

Different distributions create higher prediction error.

7. Improper Data Splitting

If the dataset is split incorrectly,

- important patterns may appear only in training,
- testing samples become significantly different.

This results in a large testing MSE.

Reasoning:

Random and representative data splitting is essential.

8. Lack of Regularization

Regularization techniques reduce model complexity.

Without regularization,

- weights become excessively large,
- the network becomes too specialized.

Reasoning:

Large weights increase overfitting.

9. Excessive Number of Training Epochs

Training for too many epochs causes the model to memorize training samples.

Initially,

- both training and testing errors decrease.

Later,

- training error continues decreasing,
- testing error begins increasing.

This is a clear sign of overfitting.

10. Hyperparameter Selection

Incorrect hyperparameters may also contribute.

Examples include

- very high learning rate,
- very low learning rate,
- large batch size,
- inappropriate optimizer,
- incorrect activation functions.

These settings affect model convergence and generalization.

Interpretation of the Given MSE Values

Training MSE = **0.85**

- Very low error.
- Indicates excellent performance on training data.

Testing MSE = **5.20**

- Much larger error.
- Indicates weak performance on unseen data.

Difference:

$$5.20 - 0.85 = 4.35$$

A difference of **4.35** between training and testing MSE is significant.

This suggests

- high variance,
- overfitting,
- poor generalization capability.

Corrective Measures**1. Apply Regularization**

Use

- L1 Regularization
- L2 Regularization (Weight Decay)

Benefits:

- Reduces large weights.
- Controls model complexity.
- Improves generalization.

2. Use Dropout

Dropout randomly deactivates neurons during training.

Benefits:

- Prevents neuron dependency.
- Reduces overfitting.
- Produces more robust models.

Typical dropout rate:

20–50%

3. Early Stopping

Monitor validation loss during training.

Stop training when validation loss starts increasing.

Benefits:

- Prevents overfitting.
- Saves training time.
- Produces better generalization.

4. Increase Training Dataset

Collect more student records.

Include

- different academic years,
- different performance levels,
- diverse student backgrounds.

Benefits:

- Better learning.
- Lower variance.
- Improved prediction.

5. Perform Data Augmentation (where applicable)

Although commonly used in images, tabular data can be improved through

- synthetic data generation,
- bootstrapping,
- data balancing.

Benefits:

- More representative dataset.
- Reduced overfitting.

6. Improve Feature Engineering

Select only important features.

Examples:

- attendance,
- assignment scores,
- internal marks,
- previous GPA,
- practical performance.

Remove irrelevant variables.

Benefits:

- Higher prediction accuracy.
- Better model understanding.

7. Reduce Model Complexity

If the network is very large, reduce

- number of hidden layers,
- neurons,
- trainable parameters.

Benefits:

- Lower variance.
- Better generalization.

8. Cross-Validation

Use K-Fold Cross Validation.

Benefits:

- Better evaluation.
- Stable performance estimate.
- Less dependence on one train-test split.

9. Normalize or Standardize Data

Scale all input features.

Methods:

- Min-Max Scaling
- Standardization (Z-score)

Benefits:

- Faster convergence.
- Stable optimization.
- Improved prediction accuracy.

10. Tune Hyperparameters

Optimize

- learning rate,
- batch size,
- optimizer,
- activation function,
- number of epochs.

Methods:

- Grid Search
- Random Search
- Bayesian Optimization

Benefits:

- Better convergence.
- Improved testing accuracy.

Expected Result After Applying Corrective Measures

After implementing these techniques:

- Training MSE may increase slightly (for example, from **0.85** to around **1.2**).
- Testing MSE should decrease significantly (for example, from **5.20** to around **1.5–2.0**).
- The gap between training and testing errors becomes much smaller.
- The model generalizes better to unseen student data.

Conclusion

The large difference between the **training MSE (0.85)** and **testing MSE (5.20)** indicates that the neural network is **overfitting** the training data. Although it performs exceptionally well on the training set, it fails to make accurate predictions on new data. The main reasons include excessive model complexity, insufficient training data, noisy data, poor feature selection, lack of regularization, and improper hyperparameter tuning. Applying corrective measures such as **dropout, regularization, early stopping, cross-validation, improved feature engineering, normalization, and hyperparameter optimization** can significantly improve the model's generalization performance, resulting in lower testing error and more reliable predictions.

Code of Conduct:

I M. Srilakshmi (192525404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Srilakshmi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

